BTC-Net V2: Spatial-Priority Hierarchical Reconstruction for Hyperspectral Image Compression

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***Abstract*—In recent years, lossy compression of hyperspectral images (HSIs) for spaceborne applications has garnered significant attention. The limited storage capacity and constrained transmission bandwidth of spaceborne equipment make it challenging to simultaneously balance the compression rate of transmitted data and the quality of locally reconstructed images. To address this challenge, we propose bit-level tensor data compression network (BTC-Net) V2—an advanced iteration of BTC-Net—that substantially enhances the performance of its predecessor. Specifically, in the encoder stage, we adopt a large-kernel convolution to replace the three-layer convolutional network used in the previous version, enabling a more lightweight encoder design that aligns with the resource constraints of spaceborne HSI compression. However, in the decoder stage, we replace the "feature enhancement followed by upsampling" paradigm of the previous version with a spatial-priority hierarchical architecture that prioritizes spatial dimension upsampling. This design ensures the subsequent feature enhancement backbone operates on a larger spatial scale—directly mitigating the loss of spatial correlations in BTC-Net and ultimately yielding higher reconstruction quality. Extensive experimental results demonstrate that our proposed BTC-Net V2 outperforms BTC-Net by 3.35 dB in PSNR—achieving 40.78 dB versus BTC-Net's 37.43 dB—while operating at an even lower bit rate (0.054 bits per pixel per band (bpppb) compared to BTC-Net's 0.060 bpppb). Our code is available at https://github.com/MWan-deeplearner/BTC-Net-V2**

***Index Terms*—High-quality reconstruction, lightweight encoder, spaceborne hyperspectral image (HSI) compression.**

I. Introduction

HYPERSPECTRAL images (HSIs) are distinguished by their ability to capture detailed spectral information with high spectral resolution, providing a significant advantage for a wide range of applications. This unique capability facilitates precise classification and target detection, making HSIs a powerful tool for advanced analysis. Consequently, research and applications leveraging HSIs have garnered substantial attention [1]. Notably, HSIs have been increasingly integrated into diverse domains, such as environmental monitoring [2], food safety assessment [3], military applications [4], anomaly detection [5], precision agriculture [6], and hyperspectral unmixing [7].

However, the acquisition and processing of HSIs are accompanied by significant challenges. The transmission bandwidth of hyperspectral imaging satellites is inherently constrained, while the high spectral resolution of HSIs results in substantial data volumes, further complicating efficient transmission. These factors collectively render the simultaneous achievement of efficiency and accuracy in HSIs transmission a critical challenge [8], [9]. As a result, the development of advanced deep compression methods specifically tailored for HSIs has become an urgent priority.

To efficiently process high-dimensional HSI data, following the compression standard of natural images, a variety of compression techniques based on frequency-domain transformations have been developed. These approaches typically utilize the discrete cosine transform (DCT) [10] or the discrete wavelet transform (DWT) [11] to concentrate the majority of the signal energy into low-frequency components. Following this transformation, less significant high-frequency components are either discarded or subjected to further compression through quantization and entropy coding techniques, such as Huffman coding and arithmetic coding [12], [13]. Beyond frequency-domain approaches, compressed sensing (CS)-based methods have also gained significant attention. CS techniques directly obtain a sparse representation through a small number of randomly sampled measurements [14], thereby integrating sampling and compression into a single step. This process effectively reduces the acquisition and storage requirements by minimizing redundant information. However, CS-based methods entail complex reconstruction procedures that demand high computational resources and advanced algorithmic optimization [15]. In addition, these methods often suffer from low reconstruction quality, which remains a critical challenge. These challenges persisted until the introduction of deep learning into the field of HSI compression. Learning-based methods typically leverage neural networks to extract features from the original HSIs and compress the data, while a second neural network at the receiving

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end reconstructs the HSI from the compressed representation. Unlike traditional frequency-domain and CS-based methods, learning-based approaches can adaptively extract complex spatial and spectral features from the original data. This data-driven framework facilitates a more efficient capture of multidimensional correlations, significantly improving compression performance. Most existing learning-based methods adopt a two-step reconstruction process: first, compressed HSI features are enhanced using a convolutional neural network (CNN) [16] or a transformer block [17]; second, the enhanced features are upsampled to restore the original spatial and spectral resolution.

Bit-level tensor data compression network (BTC-Net) [9], a learning-based method introduced in 2024, addresses the challenges of inefficient HSI transmission arising from the limited computational resources, storage capacity, and transmission bandwidth of spaceborne equipment. The architecture of BTC-Net consists of a quantized lightweight encoder and an attention-based decoder, adhering to the conventional workflow of learning-based HSIs compression methods. Specifically, the encoder is meticulously designed to be highly lightweight, substantially alleviating the computational burden on satellites, while the decoder integrates multiple layers to enhance compressed features at a small spatial scale, facilitating subsequent spatial–spectral scale restoration. Remarkably, BTC-Net achieves advanced reconstruction performance even at extremely low bit rates, PSNR 34.42 dB using an 11-bit encoder with 0.039 bits per pixel per band (bpppb). However, there is still room for improvement and enhancement as follows.

1) The encoder of BTC-Net compresses the input HSI into low-dimensional features through three successive compression stages. However, it has not yet achieved the theoretical limit of extreme lightweight design, indicating potential for further optimization.
2) The key quantization parameters, including the maximum and minimum value ranges, as well as the coefficients of fitting equations, are learned during training. However, variations in the distribution of different training datasets may complicate the learning process, making it challenging to acquire the optimal quantization parameters.
3) Since feature enhancement is performed at a low spatial scale, it fails to adequately capture spatial correlations, resulting in reconstructed HSIs with imprecise spatial details.

To overcome the limitations of existing learning-based methods, particularly BTC-Net, this article introduces BTC-Net V2, an advanced learning-based framework designed for efficient compression and high-quality reconstruction. Specifically, to tackle the first two challenges of BTC-Net, we substitute the three-layer convolutional structure in BTC-Net with a large-kernel convolution and replace BTC-Net's parameterized quantization with direct uniform quantization. This dual modification significantly reduces computational complexity and shrinks the parameter space during the encoder's convolution and quantization processes—critical for adapting to the resource-constrained on-board environment of spaceborne HSI compression. However, to address BTC-Net's third challenge, we propose a novel reconstruction paradigm: spatial upsampling is prioritized to provide the subsequent feature enhancement backbone with features of higher spatial resolution. This design enables more accurate restoration of spatial details, which is essential for preserving fine-grained information in spaceborne HSIs.

The key distinction between BTC-Net V2 and previous learning-based approaches (including BTC-Net) [8], [9]—particularly in reconstruction workflow—is illustrated in Fig. 1. Building on our earlier solution to BTC-Net's third challenge (insufficient spatial correlation capture), we replace the conventional two-stage "feature enhancement → spatial–spectral upsampling" pipeline with a three-stage hierarchical paradigm: spatial upsampling, feature enhancement (detail restoration), and spectral upsampling. This design leverages large-scale spatial representations to further boost the feature enhancement backbone's performance, as previously highlighted. The main contributions of this work are summarized as follows.

1) We propose an efficient single-layer (ESL) quantized encoder, which comprises only one convolutional layer. This design significantly alleviates the computational resource constraints of satellite systems.
2) We introduce a spatial-priority hierarchical reconstruction (SPHR) decoder that reconstructs HSIs through a three-stage sequential process: spatial resolution restoration, image detail refinement, and spectral resolution recovery. This spatial-priority hierarchical architecture significantly enhances reconstruction quality by performing feature enhancement at multiple large spatial scales.
3) Our method achieves an average PSNR of 40.93 dB using an 8-bit encoder at an extremely low compression bit rate of 0.064 bpppb, outperforming all tested SOTA methods.

II. Related Works

In this section, we present a brief overview of the HSI compression methods from the following two categories: traditional methods and learning-based methods.

A. Approaches to HSI Compression

1) Traditional Approaches to HSI Compression: Traditional approaches can be broadly categorized into two types: frequency-domain methods and CS-based methods. JPEG2000, an advanced iteration of JPEG [18], replaces the DCT [10] with the DWT [11], offering enhanced compression scalability and image quality. Du and Fowler [19] extended JPEG2000 to HSI compression. Their approach first applies principal component analysis (PCA) [20] to reduce the dimensionality of the hyperspectral data, followed by JPEG2000 compression on the dimensionally reduced HSIs. This combined method effectively balances compression efficiency and reconstruction quality for hyperspectral data.

CS-based methods are grounded in the principle that high-dimensional tensor images can be represented sparsely or nearly sparsely in specific domains [21], [22], [23]. Building

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on this foundation, Yang et al. [24] proposed SLTNCS, a HSI compression approach that leverages sparse tensor representations. Their work refined nonlinear CS theory to preserve image structures more accurately using a limited number of sensors. Subsequently, Wang et al. [25] employed tensor Tucker decomposition to capture global spatial–spectral correlations across all HSI bands. In addition, they incorporated weighted 3-D total variation (3DTV) to characterize local smooth structures in both spatial and spectral domains. To address the limitations of the assumptions inherent in 3DTV, Peng et al. [21] introduced an enhanced 3DTV (E-3DTV) CS method. Unlike traditional 3DTV, which enforces sparsity directly on gradient maps, E-3DTV applies sparsity constraints to subspace bases derived from gradient maps across all HSI bands. This approach inherently encodes both the correlations and variations among spectral bands, thereby more effectively capturing the intrinsic structure of HSI data.

2) Learning-Based Approaches to HSI Compression: The significant potential of deep learning in HSI compression has been widely acknowledged. Most contemporary HSI compression networks adopt an encoder–decoder architecture [26], leveraging the strengths of deep neural networks to achieve efficient compression and reconstruction. For instance, Hsu et al. [8] proposed a deep CS network (DCSN) for HSI compression. The lightweight encoder consists of three convolutional layers to simultaneously reduce both spatial and spectral dimensions. The decoder incorporates a feature fusion block and a feature aggregation module to enhance feature representation.

In addition, Zhou et al. [9] introduced the BTC-Net. BTC-Net employs a data-driven, lightweight quantized neural encoder to eliminate spatial redundancy, followed by a Huffman coder to further reduce coding redundancy. This two-stage bit-level compression scheme achieves exceptionally low transmission bit rates. In addition, BTC-Net integrates a channelwise attention-based feature enhancement module to facilitate the reconstruction of hyperspectral tensor data.

Further advancing the field, Zhang et al. [27] developed a CNN-transformer mixture (CTM) architecture-based CS network (CTCSN). CTCSN utilizes a nonsquare convolutional layer and CTM-like modules to construct a lightweight encoder. In the decoding stage, a multilayer CTM-based backbone enhances feature representation after compression. Finally, CTCSN performs spatial and spectral super-resolution to restore the full-resolution HSI.

Building on the framework of BTC-Net, our proposed BTC-Net V2 introduces a more efficient encoder with uniform quantization and a high-quality reconstruction decoder featuring a spatial-priority hierarchical architecture, specifically designed for spaceborne HSI compression.

B. Approaches to Quantization

Quantization methods can be broadly categorized into two fundamental paradigms based on their coupling with the training procedure: post-training quantization (PTQ) and quantization-aware training (QAT).

1) PTQ: PTQ quantizes a pretrained floating-point model into lower bit weights and/or activations without retraining. Notably, Jacob et al. [28] laid a foundational framework for uniform quantization, which includes both symmetric and asymmetric mapping strategies, and elaborated on the inference process using integer arithmetic, making it highly compatible with mainstream hardware. Later in 2024, Wei et al. [29] proposed a hybrid asymmetric quantization method based on activation redistribution, which applies asymmetric quantization to convolutional layer activations (to adapt to their asymmetric distributions) and symmetric quantization to weights (to simplify computation), achieving minimal accuracy loss (e.g., only 0.39% for GoogLeNet and 0.72% for YOLOv1) while ensuring hardware-friendliness. However, PTQ may suffer from reduced accuracy due to directly quantizing pretrained models without adapting the network to low-bit representations, which limits its applicability in scenarios requiring high-precision performance.

2) QAT: QAT incorporates quantization effects during training, allowing the model to adapt to low-bit representations, thereby achieving higher accuracy at the cost of additional training. Building upon this, Zhou et al. [9] employs a data-driven approach that maps 32-bit floating-point weights and activations to b-bit fixed-point values using a differentiable asymptotic function approximating uniform quantization, adaptively optimizing parameters via a characteristic variable β to balance compression efficiency and reconstruction performance, and is combined with Huffman coding for further bit-level compression. Later, Ding et al. [30] standardized weights to optimize information entropy, binarized them into 1-bit values $(+1/-1)$ for $32\times$ weight storage reduction and multiplication-free operations, and used an adaptive gradient estimator to alleviate gradient mismatch during training. The proposed BTC-Net V2 follows the QAT paradigm of BTC-Net while discarding the trained parameters for b-bit quantization and streamlining the quantization process.

III. Proposed Method

In this section, we introduce our deep learning-based BTC-Net V2 and provide a detailed explanation of its design and functionality.

A. Review of Our Previous BTC-Net

Since this work builds upon our previous BTC-Net method [9], it is essential to provide a brief review of its architecture. BTC-Net comprises a lightweight quantized neural encoder and an attention-based super-resolution decoder. The encoder consists of three convolutional layers that progressively reduce the dimensionality of the input HSI. Let $\mathbf{X} \in \mathbb{R}^{172\times128\times4}$ and $\hat{\mathbf{Y}} \in \mathcal{Q}^{27\times32\times1}$ be the input and output of the encoder, respectively, where $\mathcal{Q} = \{0, 1, 2, \ldots, 2^b - 1\}$, and b represents the quantization bit depth. The encoding process in BTC-Net can be formulated as follows:

$$\begin{cases} \hat{\mathbf{X}} = \text{Quantize}_{V1}\left(\mathbf{X}; l_0, u_0, c_0, b\right) \\ \mathbf{Y}^1 = \text{Quantize}_{V1}\left(\phi_{\text{LReLU}}\left(\text{QConv}\left(\hat{\mathbf{X}}\right)\right); l_1, u_1, c_1, b\right) \\ \mathbf{Y}^2 = \text{Quantize}_{V1}\left(\phi_{\text{LReLU}}\left(\text{QConv}\left(\mathbf{Y}^1\right)\right); l_2, u_2, c_2, b\right) \\ \hat{\mathbf{Y}} = \text{Quantize}_{V1}\left(\phi_{\text{LReLU}}\left(\text{QConv}\left(\mathbf{Y}^2\right)\right); l_3, u_3, c_3, b\right) \end{cases} \tag{1}$$

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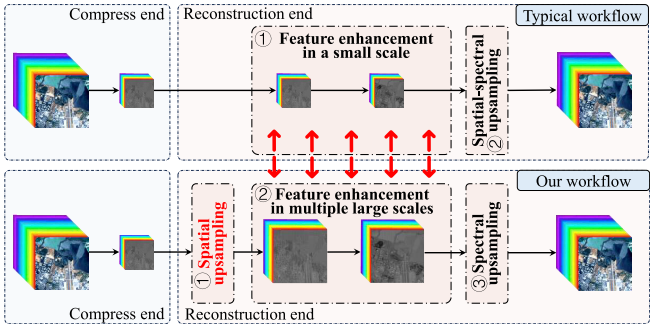


Fig. 1. Comparison of workflows between our method and previous learning-based approaches. (Top) Workflow of conventional learning-based methods for HSI compression. At the compression end, an input HSI is compressed into a low-dimensional representation. At the reconstruction end, feature enhancement on a small scale is performed first, followed by spatial and spectral super-resolution to restore the original HSI. (Bottom) Workflow of BTC-Net V2. While the compression process remains consistent with conventional methods, the reconstruction end adopts a distinct approach. Specifically, the compressed image undergoes spatial upsampling as the first step to prioritize the restoration of the spatial scale. Subsequently, feature enhancement (detail restoration) and spectral upsampling are sequentially applied to further recover the spectral scale.

where $\text{Quantize}_{V1}(\cdot;\cdot,\cdot,\cdot,\cdot)$ denotes the quantization function. The parameters l_0, l_1, l_2, l_3 and u_0, u_1, u_2, u_3 represent the lower and upper bounds for quantization, respectively, while c_0, c_1, c_2, c_3 are the coefficients of the fitting function. All of these quantization parameters are learned during training. In addition, b denotes the quantization level. However, $\phi_{\text{LReLU}}(\cdot)$ represents the leaky ReLU activation function, and $\text{QConv}(\cdot)$ denotes the parameter-quantized convolution operation. Following the quantization process, Huffman encoding and decoding are applied to further reduce the transmission bit rate.

In the decoder stage, the channelwise attention feature enhancement block (CAFE_Block) with 16 modules is first adopted to refine the compressed features in the low-dimensional scale

$$\mathbf{Y}^{\text{FE}} = \text{CAFE_Block}\left(\hat{\mathbf{Y}}\right) \tag{2}$$

where $\mathbf{Y}^{\text{FE}} \in \mathbb{R}^{27\times32\times1}$ represents the enhanced features. Subsequently, the spatial–spectral upsampling process is applied to reconstruct the image to its original size

$$\begin{cases} \mathbf{X}^2 = \phi_{\text{SiLU}}\left(\text{Conv}\left(\text{Interpolate}\left(\mathbf{Y}^{\text{FE}}\right)\right)\right) \\ \mathbf{X}^1 = \phi_{\text{SiLU}}\left(\text{Conv}\left(\mathbf{X}^2\right)\right) \\ \tilde{\mathbf{X}} = \text{Conv}\left(\phi_{\text{SiLU}}\left(\text{Conv}\left(\text{Interpolate}\left(\mathbf{X}^1\right)\right)\right)\right) \end{cases} \tag{3}$$

where $\phi_{\text{SiLU}}(\cdot)$ represents the SiLU activation function. In addition, nearest interpolation $\text{Interpolate}(\cdot)$ is used for spatial upsampling, while convolution $\text{Conv}(\cdot)$ is applied for spectral upsampling. Finally, BTC-Net reconstructs the output image $\tilde{\mathbf{X}}$ through a final convolution operation.

B. Overview of the Proposed BTC-Net V2

The BTC-Net V2 network comprises an ESL encoder and a SPHR decoder. The overall processing pipeline of BTC-Net V2 is depicted in Fig. 2. Due to the inherent imaging characteristics of hyperspectral imaging systems, full images are acquired in a strip-by-strip manner [31], [32]. Consequently, following the settings in [8] and [9], the input data are segmented into strips of size $172 \times 128 \times 4$, where 172 corresponds to the number of spectral channels, and 128×4 represents the spatial dimensions of each hyperspectral strip.

First, the HSI strips are compressed using our ESL encoder, where both the parameters and activations are quantized to minimize computational complexity and storage requirements. Subsequently, the compressed feature maps are encoded and decoded using a Huffman coding scheme to further reduce the transmission bit rate. Following decoding, the features are reconstructed through our proposed SPHR decoder to restore the original data structure. Specifically, the decoded data undergo three sequential steps: spatial upsampling in priority to restore spatial scale information, detail restoration to enhance fine-grained details, and spectral upsampling to recover spectral scale information.

C. ESL Encoder

In a typical autoencoder, the encoder compresses the input data $\mathbf{X} \in \mathbb{R}^{C_o\times H_o\times W_o}$ into a lower dimensional feature representation $\mathbf{Y} \in \mathbb{R}^{C_c\times H_c\times W_c}$. Specifically, if quantization is applied, $\hat{\mathbf{Y}} \in \mathcal{Q}^{C_c\times H_c\times W_c}$ will be the output, where $\mathcal{Q} = \{0, 1, 2, \ldots, 2^b - 1\}$, and b represents the quantization bit depth. Here, the subscript o denotes the dimensions of the original images, while c denotes the dimensions of the compressed representations.

In BTC-Net V2, the encoder is designed to be highly lightweight to minimize storage requirements and computational costs. Specifically, it employs a single convolutional layer to compress the HSIs. Furthermore, uniform quantization is applied to both model parameters and activations, effectively reducing memory consumption while maintaining efficiency.

1) Architecture of Encoder: In previous studies, researchers have employed relatively simple encoders, such as multilayer CNNs [8], [9] or attention-based modules [27], to compress HSIs. However, these approaches have yet to reach the theoretical limit of an extremely lightweight encoder.

In BTC-Net V2, we further refine the encoder design by replacing the conventional three-layer architecture [8], [9] with a streamlined single-layer convolutional network, aligning with the design choices adopted in numerous related works [33], [34], [35]. This modification effectively expands the effective receptive fields. Hyperspectral imaging equipment, particularly in spaceborne applications, operates using a strip-by-strip sensing process as mentioned earlier [31], [32]. As a result, each HSI strip tends to have a very narrow width, allowing a relatively small convolutional kernel to cover the entire width dimension. To leverage this characteristic, we directly set the kernel size of our single convolutional layer to 11×4 with 27 output channels, resulting in the feature-level compression ratio (CR) of 1%.

To improve the feature representation capability of the encoder during compression, we employ the parametric ReLU (PReLU) activation function to introduce nonlinearity to the convolutional output. The PReLU is defined as follows:

$$\phi_{\text{PReLU}}(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha x, & \text{if } x \le 0 \end{cases} \tag{4}$$

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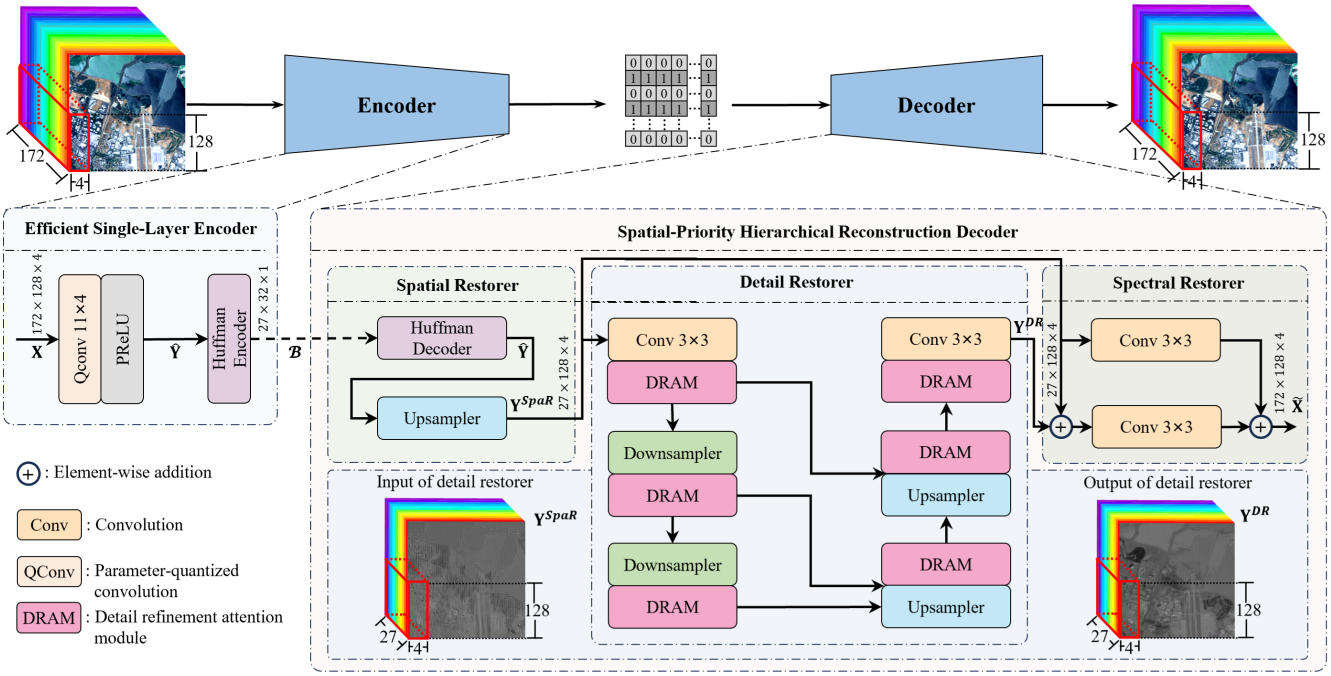


Fig. 2. Overall architecture of the proposed BTC-Net V2 method. The input HSI data undergoes initial compression using an efficient quantized single-layer encoder, generating a feature map. This feature map is then processed by a Huffman encoder–decoder to further reduce the transmission bit rate. The decoded data is reconstructed by a SPHR decoder, which restores the original data structure. The DRAM is incorporated to enhance reconstruction quality. Notably, concatenation operations between the upsampler and DRAM are ignored.

where α is a learnable parameter that can be optimized during training to better adapt to the data distribution.

Furthermore, to further minimize the compression bit rate, we incorporate a Huffman encoder [12] to eliminate the coding redundancy in the compressed HSIs. Accordingly, a Huffman decoder is introduced in the decoding phase to decode the compressed Huffman codes.

2) Quantization of Parameters and Activations: To achieve lightweight and efficient computing in the encoder, following BTC-Net [9], we quantize both the parameters and activations, thereby further reducing storage and computational resource consumption.

First, let $\boldsymbol{x} \in \mathbb{R}^D$ be the parameters or activations that need to be quantized, where the quantization level is b bits and D denotes the shape of $\boldsymbol{x}$. The range of $\boldsymbol{x}$ is then evenly divided into $2^b - 1$ intervals. Subsequently, $\boldsymbol{x}$ is mapped to the 2^b endpoints corresponding to the $2^b - 1$ intervals. Finally, during training, following the standard pipeline [28], we dequantize the quantized value $\boldsymbol{x}^q$ back to the real-number domain to maintain the floating-point representation of the result. The total pipeline can be summarized as follows:

$$\begin{cases} s = \dfrac{\max(\boldsymbol{x}) - \min(\boldsymbol{x})}{2^b - 1} \\ \boldsymbol{x}^q = \text{round}\left(\dfrac{\boldsymbol{x} - \min(\boldsymbol{x})}{s}\right) \\ \hat{\boldsymbol{x}} = \boldsymbol{x}^q \cdot s + \min(\boldsymbol{x}) \end{cases} \tag{5}$$

where s denotes the quantization scale, while b determines the number of partition intervals, thereby affecting the precision of the quantization. A smaller value of b results in reduced storage space usage but also leads to lower output accuracy. In addition, round$(\cdot)$ denotes the operation of rounding to the nearest integer. $\boldsymbol{x}^q \in \mathcal{Q}^D$ represents the discrete integer values after quantization, and $\hat{\boldsymbol{x}} \in \mathbb{R}$ represents the continuous floating-point values after dequantization, which can take on 2^b possible values, serving as the quantized approximation of the original data $\boldsymbol{x}$. The entire uniform quantization process [see (5)] can be summarized as follows:

$$\hat{\boldsymbol{x}} = \text{Quantize}(\boldsymbol{x}; b). \tag{6}$$

The model parameters $\mathbf{W}$ and $\mathbf{B}$, along with the model input $\mathbf{X}$, are quantized to improve computational efficiency, while the activations $\mathbf{Y}$ are quantized to minimize the transmission data size.

3) Total Pipeline of the Encoder: In this section, we outline the pipeline of our ESL encoder. Initially, we normalize the input HSI $\mathbf{X} \in \mathbb{R}^{C_o \times H_o \times W_o}$ to a range of $[0, 1]$ before performing convolution. Next, as illustrated in Fig. 3, we uniformly quantize both the normalized input data $\mathbf{X}$ and the parameters of the convolutional layer in the encoder. After this quantization step, we obtain b-bit fixed-point values, denoted as $\hat{\mathbf{X}}$, $\hat{\mathbf{W}}$, and $\hat{\mathbf{B}}$. Subsequently, forward propagation is performed to extract features from the quantized inputs $\hat{\mathbf{X}}$ using the quantized parameters $\hat{\mathbf{W}}$ and $\hat{\mathbf{B}}$. Finally, the output feature map also undergoes quantization to further reduce the size of the information

$$\begin{cases} \hat{\mathbf{X}} = \text{Quantize}(\mathbf{X}; b) \\ \hat{\mathbf{Y}} = \text{Quantize}\left(\phi_{\text{PReLU}}\left(\text{QConv}\left(\hat{\mathbf{X}}\right)\right); b\right) \end{cases} \tag{7}$$

where Quantize$(\cdot;\cdot)$ refers to the uniform quantization process discussed in Section III-C2. $\hat{\mathbf{Y}} \in \mathcal{Q}^{C_c \times H_c \times W_c}$ represents the quantized output.

Finally, the quantized output $\hat{\mathbf{Y}}$ is further compressed using a Huffman coder to reduce the transmission bit rate

$$\mathcal{B} = \text{HuffmanEncode}\left(\hat{\mathbf{Y}}\right) \tag{8}$$

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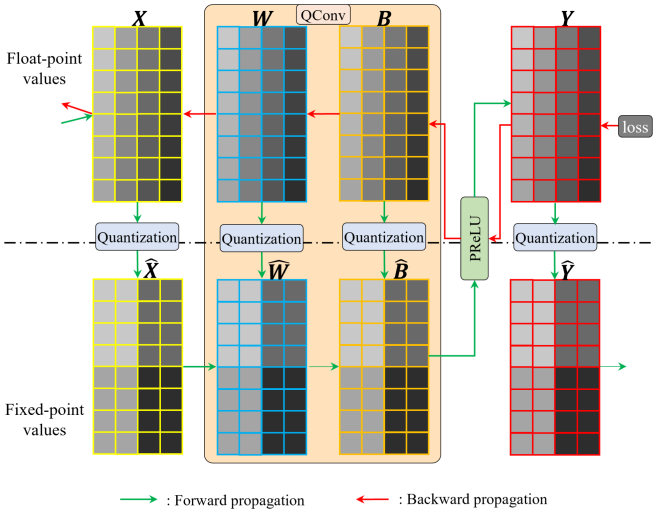


Fig. 3. Data flow of the quantized efficient encoder is as follows. First, the parameters of the convolutional layer and the input are quantized. Then, we use the quantized values to perform forward propagation. Finally, the activation is quantized to produce fixed-point values. During backward propagation, only the floating-point values are used for searching and optimization.

where HuffmanEncode(·) refers to the Huffman encoding process. $\mathcal{B}$ represents the final compressed bit stream for transmission.

The gradients of all quantized data are approximated as the gradients of their prequantized counterparts, which is accomplished using the straight-through estimation (STE) [36] algorithm to enable backward propagation. In summary, as illustrated in Fig. 3, we optimize the floating-point parameters and use their fixed-point representations for forward propagation.

D. SPHR Decoder

In this section, we introduce our SPHR decoder and highlight its advantages.

1) Overall Introduction of the SPHR Decoder: In a typical autoencoder, the decoder is responsible for decompressing or reconstructing the compressed data. Its primary objective is to restore the compressed representation as accurately as possible, ensuring a reconstruction that closely resembles the original image. Unlike conventional image super-resolution tasks [37], [38], [39], HSI compression and reconstruction involve not only spatial resolution recovery but also spectral fidelity preservation. Existing deep learning-based reconstruction methods generally follow two main processes [8], [9], [27]. First, a feature enhancement module—typically a CNN or a transformer—is employed to enhance the features of the compressed image in a small spatial scale. Second, the enhanced features are progressively upsampled, both spatially and spectrally, to restore the original image dimensions.

This two-stage pipeline may lead to the following potential issue. Since feature enhancement is performed only at a low spatial scale, it may fail to adequately capture spatial correlations, resulting in reconstructed HSIs with imprecise spatial details.

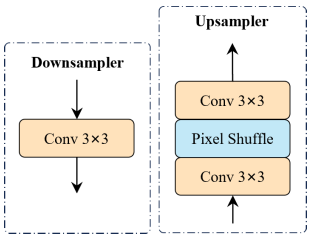


Fig. 4. Downsampler and upsampler used in spatial restorer and DR. The downsampler implements downsampling using a convolutional layer. The upsamplers introduce pixel shuffle to realize scale upsampling.

To address the aforementioned challenge, we propose a three-stage hierarchical process paradigm for reconstructing HSIs, which sequentially integrates the spatial restorer (SpaR), the detail restorer (DR), and the spectral restorer (SpeR). The core innovation of our approach lies in prioritizing spatial restoration at the initial stage. By reconstructing the full spatial scale early in the process, the restored features provide a more comprehensive and informative representation of HSIs compared to conventional methods. This enhanced information content significantly improves detail preservation, thereby optimizing the effectiveness of subsequent restoration stages.

2) Details of SPHR Decoder: We will introduce the three restoration modules and the total pipeline of the decoder in detail in this section.

a) Huffman Decoder: First, the encoded Huffman codes are decoded by a Huffman decoder

$$\hat{\mathbf{Y}} = \text{HuffmanDecode}\left(\mathcal{B}\right) \tag{9}$$

where HuffmanDecode(·) means a Huffman decoding operation corresponding to the above-mentioned Huffman encoder. In addition, $\hat{\mathbf{Y}} \in \mathbb{R}^{C_c \times H_c \times W_c}$ and $\mathcal{B}$ mean the compressed features and the encoded binary codes mentioned at Section III-C3, respectively.

b) Spatial Restorer: In our decoder, the decoded features undergo spatial restoration as the first step. The spatial restorer performs spatial upsampling using the pixel shuffle operation [40]. Pixel shuffle efficiently rearranges channel features into spatial dimensions, enabling high-quality upsampling without introducing additional interpolation errors, making it well-suited for spatial upsampling tasks.

The detailed architecture of the upsampler within the spatial restorer is illustrated in Fig. 4. The spatial restorer first employs a convolutional layer to expand the spectral dimension. Next, pixel shuffle is applied to redistribute the additional spectral information into spatial dimensions. Finally, another convolutional layer is used to further refine the spatially upsampled image. The entire pipeline can be summarized as follows:

$$\begin{cases} \mathbf{Y}^{cu} = \text{Conv}\left(\hat{\mathbf{Y}}\right) \\ \mathbf{Y}^{su} = \text{PixelShuffle}\left(\mathbf{Y}^{cu}\right) \\ \mathbf{Y}^{\text{SpaR}} = \text{Conv}\left(\mathbf{Y}^{su}\right) \end{cases} \tag{10}$$

where PixelShuffle(·) refers the pixel shuffle operation. $\mathbf{Y}^{cu} \in \mathbb{R}^{\lambda^2 C_c \times H_c \times W_c}$ represents the channel-upsampled features, and $\mathbf{Y}^{su} \in \mathbb{R}^{C_c \times \lambda H_c \times \lambda W_c}$ denotes the spatial upsampled features through pixel shuffle. Here, λ is the spatial scale factor, defined

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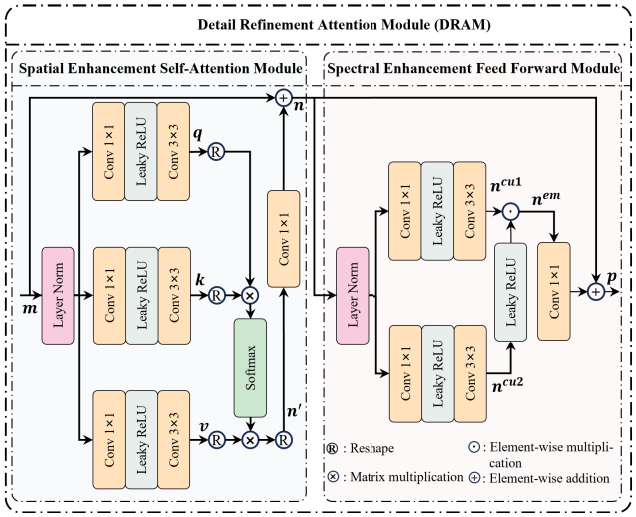


Fig. 5. DRAM in the DR. DRAM can be divided into two parts, the SESAM and the spectral enhancement feedforward module. The former implements self-attention using convolution and matrix multiplication, the latter carries out feedforward operation with elementwise multiplication.

as $\lambda = H_o/H_c = W_o/W_c$. Typically, to restore the original spatial dimensions, we set λ to 4, namely, $\lambda \times H_c = H_o$, $\lambda \times W_c = W_o$, corresponding to a 4× upsampling ratio for both the height and width scales. In addition, $\mathbf{Y}^{\text{SpaR}} \in \mathbb{R}^{C_c \times H_o \times W_o}$ is the output of the spatial restorer. After the spatial restorer, the spatial scale of the compressed features is successfully upsampled.

c) Detail Restorer: Next, we design the DR to recover the spatial and spectral details of the spatially restored images $\mathbf{Y}^{\text{SpaR}}$. The DR employs a U-shaped network architecture, a widely adopted structure in the field of image restoration [41], [42]. This U-shaped architecture enables the model to effectively fuse multiscale features, facilitating a comprehensive understanding of both local and global information within the spatially upsampled images. The core module of U-shaped DR is the detail refinement attention module (DRAM), as shown in Fig. 2.

1) *DRAM module:* The structure of DRAM is shown in Fig. 5. DRAM can be divided into two parts: the spatial enhancement self-attention module (SESAM) and the spectral enhancement feedforward module (SEFFM). Assume that $\boldsymbol{m} \in \mathbb{R}^{C \times H \times W}$ represents the input of the SESAM module. First, we normalize $\boldsymbol{m}$. This process can stabilize the numerical range and prevent gradient explosions or vanishing. Then, we obtain the query ($\boldsymbol{q} \in \mathbb{R}^{HW \times C}$), key ($\boldsymbol{k} \in \mathbb{R}^{C \times HW}$), and value ($\boldsymbol{n} \in \mathbb{R}^{HW \times C}$)

$$\boldsymbol{q}, \boldsymbol{k}, \boldsymbol{v} = \text{Reshape}\left(\text{Conv}\left(\phi_{\text{LReLU}}\left(\text{Conv}\left(\boldsymbol{m}\right)\right)\right)\right) \tag{11}$$

where Reshape(·) operation reshapes the outputs of the convolutions. In detail, $\boldsymbol{q}$ represents the regions of interest in the image. $\boldsymbol{k}$ represents the available References to different image regions. $\boldsymbol{v}$ holds the pixel value information in certain areas, retrieved based on $\boldsymbol{q}$–$\boldsymbol{k}$ matching. Ultimately, we use a convolution and a skip connection to form the output $\boldsymbol{n} \in \mathbb{R}^{C \times H \times W}$ as follows:

$$\begin{cases} \boldsymbol{n}' = \text{Reshape}\left(\phi_{\text{Softmax}}\left(\boldsymbol{q} \otimes k\right) \otimes \boldsymbol{v}\right) \\ \boldsymbol{n} = \text{Conv}\left(\boldsymbol{n}'\right) + \boldsymbol{m} \end{cases} \tag{12}$$

where $\otimes$ denotes the matrix multiplication, which is performed among $\boldsymbol{q}$, $\boldsymbol{k}$, and $\boldsymbol{v}$ to enhance pixel-level features of specific areas in the image. In addition, ϕ_{Softmax} means the softmax operation, which converts scores into probabilities, emphasizing key information. The SESAM calculates the relationship between different parts to emphasize important local information and long-range dependencies. Thus, it can enhance the spatial information in images.

For enhancing the spectral information in $\boldsymbol{n}$, we introduce the SEFFM. Specifically, the normalized $\boldsymbol{n}$ undergoes spectral upsampling to exploit improved channelwise correlations through elementwise multiplication. Subsequently, the features are spectrally downsampled to restore the original channel dimensions. These operations effectively enhance the channelwise information, thereby improving the overall feature representation. The output of SEFFM can be described as follows:

$$\begin{cases} \boldsymbol{n}^{cu_1} = \text{Conv}\left(\phi_{\text{LReLU}}\left(\text{Conv}\left(\boldsymbol{n}\right)\right)\right) \\ \boldsymbol{n}^{cu_2} = \text{Conv}\left(\phi_{\text{LReLU}}\left(\text{Conv}\left(\boldsymbol{n}\right)\right)\right) \\ \boldsymbol{n}^{\text{em}} = \boldsymbol{n}^{cu_1} \odot \phi_{\text{LReLU}}\left(\boldsymbol{n}^{cu_2}\right) \\ \boldsymbol{p} = \text{Conv}\left(\boldsymbol{n}^{\text{em}}\right) + \boldsymbol{n} \end{cases} \tag{13}$$

where $\odot$ represents the elementwise multiplication, $\boldsymbol{n}^{cu_1}$, $\boldsymbol{n}^{cu_2} \in \mathbb{R}^{\gamma C \times H \times W}$ are intermediate channel-upsampled variables, and $\boldsymbol{n}^{\text{em}} \in \mathbb{R}^{\gamma C \times H \times W}$ denotes the result of the elementwise multiplication. Besides, $\boldsymbol{p} \in \mathbb{R}^{C \times H \times W}$ is the output of SEFFM module. By upsampling the spectrum, the spectral dimension can be smoother, and the redundant information across bands can be further increased. Specifically, the channel scale of the input images is first expanded and then compressed. This "expansion–compression" mechanism enables the network to more effectively capture long-range dependencies between spectral channels and enhances its ability to fuse information across the spectral domain.

By leveraging the SESAM and SEFFM modules, DRAM enhances feature representation across both spatial and spectral dimensions. Equations (11)–(13) can be briefly expressed as follows:

$$\boldsymbol{p} = \text{DRAM}\left(\boldsymbol{m}\right). \tag{14}$$

2) *Overall Pipeline of DR:* The output of SpaR $\mathbf{Y}^{\text{SpaR}}$ is input into our DR module. Three DRAMs and two downsampling convolutions are performed to gain multiscale features as follows:

$$\begin{cases} \mathbf{F}^{d1} = \text{DRAM}\left(\text{Conv}\left(\mathbf{Y}^{\text{SpaR}}\right)\right) \\ \mathbf{F}^{d2} = \text{DRAM}\left(\text{Conv}\left(\mathbf{F}^{d1}\right)\right) \\ \mathbf{F}^{\text{mid}} = \text{DRAM}\left(\text{Conv}\left(\mathbf{F}^{d2}\right)\right) \end{cases} \tag{15}$$

where $\mathbf{F}^{d1} \in \mathbb{R}^{C_f \times H_o \times W_o}$, $\mathbf{F}^{d2} \in \mathbb{R}^{2C_f \times (H_o)/2 \times (W_o)/2}$, and $\mathbf{F}^{\text{mid}} \in \mathbb{R}^{4C_f \times (H_o)/4 \times (W_o)/4}$ are the intermediate features,

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and C_f denotes the channels in feature space. The downsampler in Fig. 2 is designed by a convolutional layer, as shown in Fig. 4.

Then, we upsample the most compressed features $\mathbf{F}^{\text{mid}}$ and fuse multiscale features between the downsampling and upsampling process. The detailed architecture of the upsampler in DR is shown in Fig. 4. Finally, the fully fused and enhanced features are transformed back to the input shape to form the output of DR, $\mathbf{Y}^{\text{DR}} \in \mathbb{R}^{C_c \times H_o \times W_o}$. The total process can be summarized as follows:

$$\begin{cases} \mathbf{F}^{\text{mid}\to u2} = \text{Conv}\left(\text{PexelShuffle}\left(\text{Conv}\left(\mathbf{F}^{\text{mid}}\right)\right)\right) \\ \mathbf{F}^{u2} = \text{DRAM}\left(\text{Conv}\left(\text{Cat}\left(\mathbf{F}^{d2}, \mathbf{F}^{\text{mid}\to u2}\right)\right)\right) \\ \mathbf{F}^{u2\to u1} = \text{Conv}\left(\text{PexelShuffle}\left(\text{Conv}\left(\mathbf{F}^{u2}\right)\right)\right) \\ \mathbf{F}^{u1} = \text{DRAM}\left(\text{Cat}\left(\mathbf{F}^{d1}, \mathbf{F}^{u2\to u1}\right)\right) \\ \mathbf{Y}^{\text{DR}} = \text{Conv}\left(\text{DRAM}\left(\mathbf{F}^{u1}\right)\right) + \mathbf{Y}^{\text{SpaR}} \end{cases} \tag{16}$$

where $\mathbf{F}^{\text{mid}\to u2}$, $\mathbf{F}^{u2} \in \mathbb{R}^{2C_f \times (H_o)/2 \times (W_o)/2}$, $\mathbf{F}^{u2\to u1} \in \mathbb{R}^{C_f \times H_o \times W_o}$, and $\mathbf{F}^{u1} \in \mathbb{R}^{2C_f \times H_o \times W_o}$ are the intermediate features. $\text{Cat}(\cdot,\cdot)$ means the channelwise concatenation. The DR effectively restores both the spatial and spectral details of the feature, enhancing the detail initiated by the preceding spatial restorer. In addition, it provides comprehensive spectral information to support the subsequent spectral restorer. As a result, the spectral restoration process can be completed efficiently in just a few steps.

d) Spectral Restorer: Finally, the spectral information will be restored as follows:

$$\tilde{\mathbf{X}} = \text{Conv}\left(\mathbf{Y}^{\text{DR}}\right) + \text{Conv}\left(\mathbf{Y}^{\text{SpaR}}\right) \tag{17}$$

where the convolution operation completes the spectral upsampling here. In addition, $\tilde{\mathbf{X}}$ is the final output of the decoder. By upsampling the output results of the spatial restorer and DR modules and subsequently aggregating them, the interference and overfitting issues during model training could be significantly mitigated. Furthermore, this approach enhances the stability and robustness of the model, ensuring more reliable performance.

E. Loss Function

On the encoding side, we employ a neural network to compress HSIs into low-dimensional tensors. Subsequently, on the decoding side, these tensors are reconstructed back into the original HSIs. Throughout this process, we utilize the $L1$ loss function to quantify the distortion introduced by the compression and reconstruction stages, ensuring the fidelity of the recovered images

$$\mathcal{L} = \frac{1}{C_o \times H_o \times W_o} \sum_{c=1}^{C_o} \sum_{h=1}^{H_o} \sum_{w=1}^{W_o} \left|\left|\mathbf{X}_{c,h,w} - \tilde{\mathbf{X}}_{c,h,w}\right|\right|_1 \tag{18}$$

where C_o, H_o, and W_o are channels, height, and width of the original HSI, respectively. This loss function describes the error between the reconstructed image and the original image through the $L1$ distance.

IV. Experimental Results

In this section, we will introduce our experimental results to show the effectiveness of our proposed BTC-Net V2.

A. Experimental Setup

1) Dataset Overview: In our work, the AVIRIS dataset[1] is used to train and test the proposed method. These data were collected by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The sensor collects 224 continuous bands from 400- to 2500-nm wavelengths. The original 224 bands of the AVIRIS dataset include some bands affected by water vapor absorption and instrument noise. To improve data quality and reduce redundancy, these bands are typically removed, finally retaining 172 bands. In addition, all negative values are cut to 0. Finally, these HSIs are cropped into 1183 subimages, each of which has the shape of $172 \times 256 \times 256$.

Since the AVIRIS dataset collects data from a wide range of regions, it is possible to measure the performance of the method on a variety of different regions. In general, these regions can be divided into four types: urban type represented by *urban* (U), field type represented by *farm* (F), mountain type represented by *mountain* (M), and lake type represented by *lake* (L). There are 101 subimages of the *urban* type, 265 subimages of the *farm* type, 707 subimages of the *mountain* type, and 110 subimages of the *lake* type.

We divide each type of subimage into a training set and a validation set in a ratio of 10:1, and finally extract five subimages from the validation set as the final test set. These five subimages are of the *urban* type, the *farm* type, the *mountain* type, the *lake* type, and the Moffett type, respectively.

2) Training Hyperparameters: We use a GeForce RTX 3090 to train our model. When training, the batch size is set to 16 and the learning rate is set to 10^{-4} at the beginning of the training process and finally decreased to 10^{-5} when the model is almost fitting. For quantized models, we first pretrain the model on 32-bit floating-point parameters, and then fine-tune the pretrained 32-bit model with b-bit fixed-point parameters to obtain quantized models, which makes the model parameters aware of quantization [28]. Considering that the hyperspectral imaging device acquires spatial information strip-by-strip and forms a complete HSI cube as the platform moves or the target moves [31], [32], the HSIs are further cropped in $172 \times 128 \times 4$ strips during training.

3) Testing Metrics: In our work, peak signal-to-noise ratio (PSNR), root mean square error (RMSE), and spectral angle mapper (SAM) are used to evaluate the reconstruction performance. In addition, bpppb is chosen to evaluate the compression performance.

The calculation of PSNR can be expressed as follows:

$$\text{PSNR}\left(\mathbf{X}, \tilde{\mathbf{X}}\right) = \frac{1}{C} \sum_{c=1}^{C} \left(10 \cdot \log_{10}\left(\frac{\text{MAX}_c^2}{\text{mse}_c}\right)\right) \tag{19}$$

where

$$\text{MAX}_c = \max_{h\in[1,H], w\in[1,W]} \left(\mathbf{X}_{c,h,w}\right)$$

and

$$\text{mse}_c = \frac{1}{H \times W} \sum_{h=1}^{H} \left(\sum_{w=1}^{W} \left(\left(\mathbf{X}_{c,h,w} - \tilde{\mathbf{X}}_{c,h,w}\right)^2\right)\right)$$

[1] https://aviris.jpl.nasa.gov/dataportal/

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TABLE I
QUANTITATIVE COMPARISON BETWEEN SOTA METHODS UNDER 1% FEATURE COMPRESSION RATE

Methods \ Test Data	HyspIRI17(F)	HyspIRI40(M)	California(U)	FtMyers(L)	Moffett	Average	bpppb↓
	PSNR (dB) ↑ /RMSE ↓ /SAM (°) ↓						
SLTNCS [24] (32 bit)	32.21/187.8/5.236	28.60/177.2/4.399	30.71/233.5/6.433	27.90/451.7/14.96	28.27/430.1/11.68	29.53/296.1/8.543	0.314
PCA+JPEG2000 [19](8 bit)	28.86/143.9/5.068	32.87/43.36/2.287	29.43/162.6/5.844	28.07/331.3/9.929	29.20/307.8/8.813	29.69/197.8/6.388	0.064
E3DTV [21] (32 bit)	28.42/137.3/3.609	35.18/69.91/1.751	31.95/164.4/4.663	30.99/288.9/7.821	28.84/266.9/6.772	31.08/185.5/4.923	0.314
JtenRe3DTV [25] (32 bit)	27.40/326.6/11.45	31.64/124.8/3.223	31.28/218.6/5.403	28.01/445.7/12.30	29.65/366.9/8.999	29.60/296.5/8.276	0.314
DCSN [8] (32 bit)	37.06/57.98/1.385	38.85/25.47/0.773	33.95/89.59/2.239	34.34/126.9/2.272	34.46/148.4/3.226	35.73/89.66/1.979	0.314
BTC-Net [9] (32 bit)	38.25/49.04/1.454	40.82/18.92/0.811	36.08/65.97/2.318	36.62/89.81/2.451	37.80/105.1/3.236	37.91/65.78/2.054	0.314
CTCSN [27] (32 bit)	40.27/37.81/1.271	42.59/15.35/0.776	38.39/53.30/1.952	38.81/73.67/1.768	39.00/95.64/3.085	39.81/55.15/1.770	0.314
BTC-Net V2 (32 bit)	**43.16/27.19/0.992**	**44.67/12.68/0.644**	**39.33/47.89/1.774**	**40.06/65.71/1.591**	**39.72/88.33/2.865**	**41.39/48.36/1.573**	0.314
BTC-Net V2 (8 bit)	**42.69/28.13/1.030**	43.67/**13.61/0.690**	**38.98/48.14/1.805**	**39.81/65.16/1.668**	**39.50/88.25/2.875**	**40.93/48.66/1.614**	0.064
BTC-Net V2 (7 bit)	42.67/28.74/1.035	**43.68**/14.17/0.697	38.87/50.38/1.851	39.42/70.75/1.786	39.26/91.98/2.947	40.78/51.21/1.663	0.054
BTC-Net V2 (6 bit)	41.96/31.28/1.114	42.35/16.47/0.764	38.45/52.60/1.911	39.06/73.72/1.888	38.77/96.85/3.082	40.12/54.18/1.752	0.044
BTC-Net V2 (5 bit)	40.37/37.73/1.289	40.31/20.72/0.889	37.39/58.87/2.079	38.21/82.19/2.123	37.51/108.3/3.357	38.76/61.57/1.947	**0.034**
BTC-Net V2 (4 bit)	37.71/52.85/1.701	36.88/30.77/1.165	35.39/74.48/2.448	37.71/102.1/2.526	35.58/134.5/4.074	36.41/78.95/2.383	**0.025**

where H, W, and C represent the height, width, and channels of the HSI. $\mathbf{X}$ and $\tilde{\mathbf{X}}$ are the original HSI and the reconstructed HSI, respectively.

Then, RMSE can be described as

$$\mathrm{RMSE}\left(\mathbf{X},\tilde{\mathbf{X}}\right)=\sqrt{\frac{1}{C}\sum_{c=1}^{C}\mathrm{mse}_c}. \tag{20}$$

SAM is computed as follows:

$$\mathrm{SAM}\left(\mathbf{X},\tilde{\mathbf{X}}\right)=\frac{1}{H\times W}\sum_{h=1}^{H}\left(\sum_{w=1}^{W}\arccos\left(\frac{\mathrm{dot}_{h,w}}{\mathrm{norm}_{h,w}}\right)\right) \tag{21}$$

where

$$\mathrm{dot}_{h,w}=\sum_{c=1}^{C}\left(\mathbf{X}_{c,h,w}\cdot\tilde{\mathbf{X}}_{c,h,w}\right)$$

and

$$\mathrm{norm}_{h,w}=\sqrt{\sum_{c=1}^{C}\mathbf{X}_{c,h,w}^2\cdot\sum_{c=1}^{C}\tilde{\mathbf{X}}_{c,h,w}^2}.$$

Finally, bpppb can be represented as

$$\mathrm{bpppb}=\frac{\mathrm{Length}\left(\mathcal{B}\right)}{C\times H\times W} \tag{22}$$

where Length($\mathcal{B}$) means the total transmission bits after compression and encoding. It can be computed as follows:

$$\mathrm{Length}\left(\mathcal{B}\right)=\mathrm{Length}\left(\mathcal{B}^{\hat{\mathbf{Y}}}\right)+\mathrm{Length}\left(\mathcal{B}^{H}\right)$$

where $\mathcal{B}^{\hat{\mathbf{Y}}}$ and $\mathcal{B}^{H}$ are the compressed bit streams of encoder output $\hat{\mathbf{Y}}$ and Huffman frequency table accordingly, respecctively. It should be noted that in this article, we have made a correction to the calculation method of bpppb. In the BTC-Net [9], the calculation of bpppb did not include $\mathcal{B}^{H}$. In this article, we have added this item.

B. Compression Quality Comparison With SOTA Methods

In this section, we compare our method with other SOTA approaches through both quantitative and qualitative evaluations.

1) Quantitative Comparison: The bit rate of compressed image transmission and the quality of reconstructed images have long been central concerns in HSI compression. To quantitatively assess the compression and reconstruction performance of our proposed method, we conducted a comprehensive comparative experiment, benchmarking it against seven SOTA HSI compression methods: SLTNCS [24], PCA + JPEG2000 [19], E3DTV [21], JtenRe3DTV [25], DSCN [8], BTC-Net [9], and CTCSN [27].

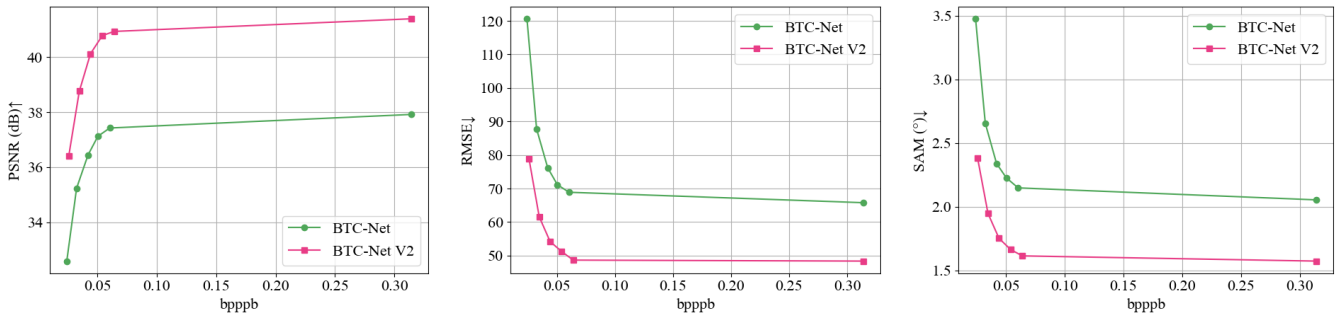
To ensure a focused comparison of reconstruction quality and transmission bit rate, we constrained the feature CR of all methods (except PCA + JPEG2000) to 1%. This corresponds to compressing the original data from a shape of 172 × 128 × 4 to a compressed shape of 27 × 32 × 1. Specifically, for PCA + JPEG2000, the transmission bit rate was set to 0.064 to match that of BTC-Net V2 (8 bit), enabling a fair evaluation of the quantized model's performance.

It is important to note that, during the transmission process, all comparative methods—except for our proposed BTC-Net V2 and the PCA + JPEG2000 method—operate with 32-bit floating-point precision. In contrast, the PCA + JPEG2000 method adopts a default 8-bit quantization scheme (consistent with its standard implementation) for data transmission.

The quantitative results are presented in Table I, where the best results are highlighted in red and the second-best results in blue. From these results, we can observe as follows.

1) BTC-Net V2 (32-bit) demonstrates superior performance across all evaluation metrics compared to the benchmarked methods. Specifically, it achieves a PSNR of 41.39 dB, surpassing the second-best method, CTCSN (39.81 dB), by a significant margin of 1.58 dB.

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(a) (b) (c)

Fig. 6. Detailed comparison between BTC-Net and BTC-Net V2. The tested metrics are (a) PSNR, (b) RMSE, and (c) SAM. Both methods are trained on 1% feature CR. The experimental results show that BTC-Net V2 outperforms BTC-Net in every testing dimension. All results in this figure are mean values of five tested HSI cubes.

Furthermore, BTC-Net V2 attains an RMSE of 48.36, which is 6.79 lower than that of CTCSN (55.15). In addition, BTC-Net V2 achieves a SAM of 1.573°, outperforming CTCSN (1.770°) by 0.197°. These results highlight the exceptional reconstruction quality of our proposed method.

2) BTC-Net V2 (8 bit) demonstrates superior performance across all three evaluation metrics compared to the benchmarked methods. Specifically, it achieves a PSNR of 40.93 dB, surpassing the second-best method, CTCSN (39.81 dB), by a significant margin of 1.12 dB. In addition, BTC-Net V2 (8 bit) attains an RMSE of 48.66, which is 6.49 lower than that of CTCSN (55.15), and a SAM of 1.614°, outperforming CTCSN (1.770°) by 0.156°. Furthermore, at the same bit rate of 0.064 bpppb, BTC-Net V2 (8 bit) significantly outperforms PCA + JPEG2000, with an improvement of 11.24 dB in PSNR, a reduction of 149.14 in RMSE, and a decrease of 4.774° in SAM. Notably, these results underscore that BTC-Net V2 (8 bit) achieves exceptional reconstruction quality even at an extremely low bit rate, demonstrating its capability to balance high performance with minimal transmission requirements.
3) BTC-Net V2 (4 bit) could outperform full precision methods (SLTNCS, PCA+JPEG2000, E3DTV, JtenRe3DTV, and DCSN) in reconstruction metrics, such as PSNR and RMSE, even at an extremely low bit rate of 0.025 bpppb. This result highlights the potential of BTC-Net V2 to effectively reconstruct HSIs under extremely low bit rate conditions.
4) BTC-Net V2 demonstrates superior performance compared to other learning-based methods, including DCSN [8], BTC-Net [9], and CTCSN [27]. This significant improvement highlights the effectiveness of our proposed SPHR framework, which enhances features across multiple large spatial scales, in contrast to those methods limited to small-scale feature enhancement.

In addition, we conducted a detailed comparison between BTC-Net V2 and BTC-Net. In this experiment, we set the feature compression rate to 1% and evaluated quantized models ranging from 4- to 8-bit precision. The experimental results, as presented in Fig. 6, reveal the following findings.

1) BTC-Net V2 is far superior to its predecessor, BTC-Net. The PSNR comparison curve of BTC-Net V2 consistently remains above that of BTC-Net, while its RMSE and SAM curves are consistently lower. This demonstrates that BTC-Net V2 achieves higher quality image reconstruction under the same transmission bandwidth. Furthermore, it significantly reduces the transmission bit rate while maintaining comparable reconstruction quality. These results underscore the comprehensive superiority of BTC-Net V2 over its predecessor. Specifically, the advantage can be attributed to the three-stage SPHR decoder, which enhances reconstruction quality by prioritizing spatial resolution restoration, followed by detail refinement, and spectral recovery.
2) In terms of the three evaluation metrics, both BTC-Net and the proposed BTC-Net V2 initially exhibit a rapid increase in PSNR, accompanied by a corresponding decrease in RMSE and SAM as the bpppb increases. However, as the bpppb continues to rise, the rate of improvement gradually slows and eventually stabilizes. Notably, both models achieve their optimal reconstruction performance at 32-bit quantization.

2) Qualitative Comparison: To assess the capability of the proposed method in preserving HSI details, we visualize the testing results. The comparison includes five methods: BTC-Net V2 (8 bit), CTCSN [27], BTC-Net [9], DCSN [8], and PCA + JPEG2000 [19]. Given that our HSIs contain 172 spectral bands—far exceeding the number in conventional optical images—we select the 25th, 15th, and 6th bands to represent the red (R), green (G), and blue (B) channels in the visualized images. In addition, we zoomed in on a small region within each image to further assess the effectiveness of detail preservation. Since the hyperspectral data was cropped to a size of $172 \times 128 \times 4$ during training, the compressed and reconstructed images were subsequently stitched together to form a size of $172 \times 256 \times 256$.

The qualitative comparison results are presented in Fig. 7. From these results, we observe that the proposed BTC-Net

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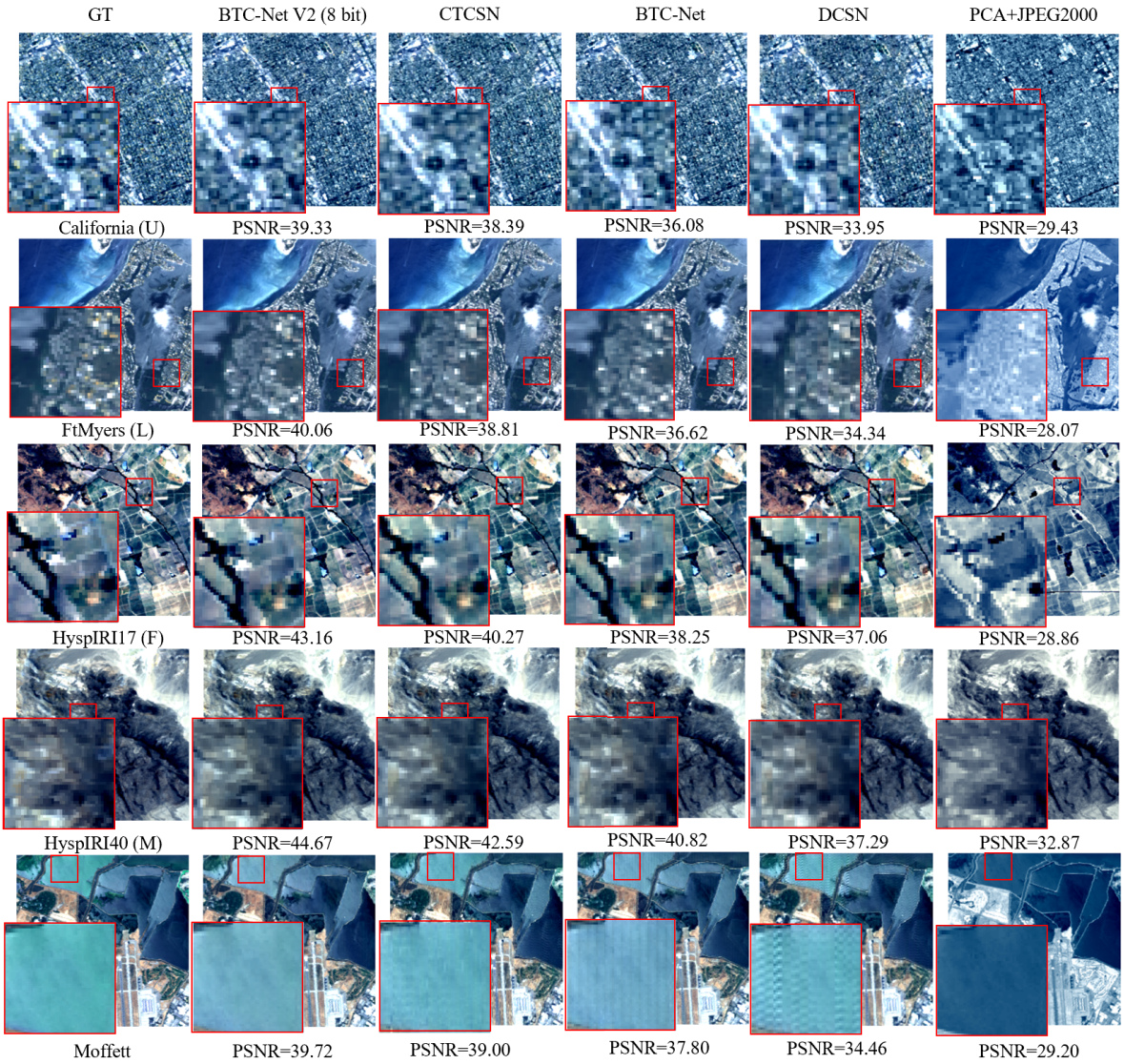


Fig. 7. Qualitative comparison of the proposed method with SOTA approaches. The comparison includes learning-based methods CTCSN [27], BTC-Net [9], DCSN [8], and our BTC-Net V2 (8 bit), as well as the frequency-domain method PCA + JPEG2000 [19]. Small red boxes indicate regions of interest in the full-size images, while large red boxes display the corresponding enlarged regions. The PSNR value for each method is also provided beneath the respective images for reference.

V2 demonstrates superior performance compared to all benchmarked methods from a visualization perspective. Specifically, as follows.

1) On the Moffett test image, the reconstruction produced by our method exhibits significantly fewer edge artifacts during stitching compared to CTCSN, BTC-Net, and DCSN. This improvement may be attributed to our SPHR decoder, which restores the spatial scale of the image at an early stage in the reconstruction process. By doing so, subsequent feature enhancement is applied at multiple large scales, effectively mitigating edge mismatches. In contrast, other methods perform feature enhancement on a smaller scale and defer all upsampling steps until the final stage, resulting in more pronounced edge inconsistencies.
2) On test images California, FtMyers, and HyspIRI40, the proposed method consistently outperforms all comparison methods in terms of image detail preservation, exhibiting fewer large blurred areas. This improvement may be primarily attributed to our DR, which effectively fuses multiscale information during the process of continuous downsampling and upsampling. By leveraging this approach, the method successfully restores both spatial and spectral details of the image, yielding a more refined and visually accurate reconstruction.

C. Complexity Analysis

In this section, we analyze the complexity and efficiency of both the proposed ESL encoder and the overall architecture.

1) Comparison of Encoder Efficiency: As previously discussed, computational efficiency is a critical factor for satellites equipped with hyperspectral imaging systems. It determines whether the satellite can compress and transmit

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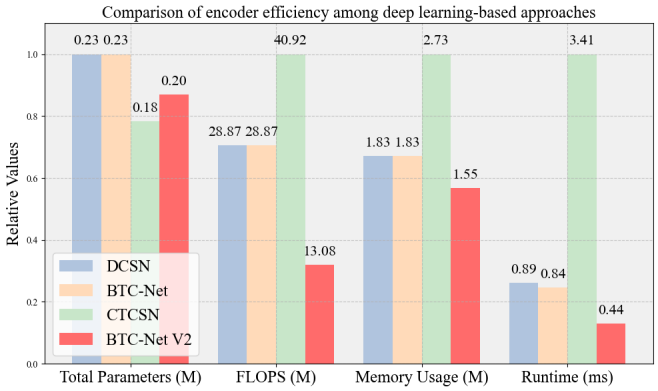


Fig. 8. Comparison of encoder efficiency is presented, including the methods DCSN [8], BTC-Net [9], CTCSN [27], and the proposed BTC-Net V2. The comparison focuses on total parameters, FLOPs, memory usage, and running time. The vertical axis of the figure presents the relative values for these metrics. In addition, the absolute values are indicated on each bar corresponding to the respective method.

the image immediately after the imaging device captures the image strip, which is essential for enabling efficient information exchange between the satellite's sender and the ground-based receiver.

In this article, we propose an ESL encoder aimed at enhancing the computational efficiency of the compression process. To evaluate the effectiveness of this encoder in reducing computational load and storage requirements on the compression side, we compare the proposed BTC-Net V2 (32-bit) with three other deep learning-based methods: DCSN [8], BTC-Net [9], and CTCSN [27]. The comparison is conducted using metrics such as total parameters, floating-point operations (FLOPs), memory usage, and running time, with all methods evaluated on a GeForce RTX 3090 GPU.

The results are presented in Fig. 8. Our BTC-Net V2 exhibits the second-lowest number of parameters (0.20 M). Furthermore, the FLOPs of our method (13.08 G) are significantly lower than those of the other comparison methods, being less than half of the second-lowest DCSN (28.97 G). In addition, the memory usage of our method (1.55 M) is lower than all the benchmarked methods. Furthermore, the running time of our method (0.44 ms) is nearly half that of the second-fastest method, BTC-Net (0.84 ms). We also evaluate the quantized version of BTC-Net V2 with an 8-bit quantization on CPU, which further reduces the running time to just 0.38 ms.

In conclusion, our method demonstrates superior encoder efficiency compared to all benchmarked methods. This is primarily attributed to the proposed single-layer architecture, which utilizes a single convolutional layer to enhance encoder efficiency while maintaining robust feature extraction ability.

2) Comparison of Overall Efficiency: To further evaluate the overall running efficiency of our method, we compare the total running time of BTC-Net V2 (32-bit) with seven other compression methods on a NVIDIA GeForce RTX 3090. In addition, we also consider bpppb and PSNR. All metrics are estimated using the mean values derived from the five tested HSIs.

TABLE II
COMPARISON ON RUNTIME OF DIFFERENT METHODS

Methods	Runtime ↓	bpppb↓	PSNR (dB)↑
PCA+JPEG2000 (8 bit) [19]	1.51 s	**0.064**	29.69
SLTNCS (32 bit) [24]	108.77 s	0.314	29.53
JtenRe3DTV (32 bit) [25]	993.60 s	0.314	29.60
E3DTV (32 bit) [21]	752.87 s	0.314	31.08
DCSN (32 bit) [8]	**17.72 ms**	0.314	35.73
BTC-Net (32 bit) [9]	20.86 ms	0.314	37.91
CTCSN (32 bit) [27]	58.64 ms	0.314	**39.81**
BTC-Net V2 (32 bit)	**6.85 ms**	0.314	**41.39**

The results are presented in Table II, with the best results highlighted in red and the second-best results highlighted in blue. From the table, we observe that the overall running time of our method (32-bit) is the shortest (6.85 ms), being 10.87 ms shorter than the second fastest method, DCSN (17.72 ms), while achieving a PSNR improvement of 5.66 dB. In addition, the second-highest PSNR (39.81 dB) is achieved by CTCSN, which requires 51.79 ms more than the proposed BTC-Net V2. However, its PSNR is 1.58 dB lower than that of BTC-Net V2. These results highlight the efficiency of the proposed method.

D. Ablation Study

1) Holistic Ablation of Core Model Components: In this section, we present the results of the ablation study to validate the effectiveness of each proposed component of our methods. In this study, we propose an ESL encoder and an SPHR decoder to enhance reconstruction quality while maintaining an extremely low bit rate. In addition, a simple uniform quantizer is introduced to further reduce computational complexity and model parameters.

To evaluate the effectiveness of these components, we tested the performance of BTC-Net V2 with and without them. The results are presented in Table III. Specifically, the baseline in the table refers to BTC-Net [9] (32 bit), a SOTA method, and the previous version of our model. For clarity, we denote BTC-Net as V1 and the proposed method as V2 throughout the left parts of this article. Specifically, for the encoder, V1 employs a three-layer convolutional architecture, while V2 adopts a streamlined single-layer design. For the decoder, V1 utilizes a two-stage decoder that performs feature enhancement at a small scale, whereas V2 introduces a three-stage SPHR decoder that conducts feature enhancement across multiple larger scales. For the quantizer, V1 incorporates a complex quantizer with three learnable parameters during training, while V2 directly implements a simple uniform quantizer.

For these compared methods, the baseline-I comprises V1's encoder and V2's decoder, whereas baseline-II consists of V1's decoder and V2's encoder. Baseline-III represents an 8-bit quantized version of BTC-Net V1. Baseline-IV retains V1 but applies V2's quantization method for model parameters

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TABLE III
ABLATION EXPERIMENTS ON COMPONENT EFFECTIVENESS OF THE PROPOSED METHOD

	V1 Encoder	V2 Encoder	V1 Decoder	V2 Decoder	V1 Quantizer (8bit)	V2 Quantizer (8bit)	PSNR ↑ /RMSE ↓ /SAM ↓	bpppb↓
Baseline	✓		✓				37.91/65.78/2.054	0.314
Baseline-I	✓			✓			41.04/49.48/1.647	0.314
Baseline-II		✓	✓				38.97/54.46/1.881	0.314
Baseline-III	✓		✓		✓		37.43/68.89/2.148	**0.060**
Baseline-IV	✓		✓			✓	37.85/58.52/1.762	**0.063**
Baseline-V		✓		✓	✓		40.52/50.21/1.701	0.064
Ours (32 bit)		✓		✓			**41.39/48.36/1.573**	0.314
Ours (8 bit)		✓		✓		✓	**40.93/48.66/1.614**	0.064

and activations. Conversely, baseline-V corresponds to V2 but utilizes V1's quantization method.

For reference, we also include both the 32-bit floating-point and 8-bit quantized versions of BTC-Net V2 at the bottom of the table. The evaluation metrics include PSNR, RMSE, SAM, and bpppb, with the best results highlighted in red and the second-best in blue. The experimental results, summarized in Table III, demonstrate that as follows.

1) *Effectiveness of Our Single-Layer Encoder:* The comparison between baseline-I and Ours (32 bit) reveals that removing our ESL encoder leads to a PSNR decrease of 0.35 dB, an RMSE increase of 1.12, and a SAM increase of 0.074°. These variations highlight the effectiveness of the proposed ESL encoder. In addition, the comparison between baseline and baseline-II shows that replacing V1's encoder with V2's encoder improves V1's performance, resulting in a PSNR increase of 1.06 dB, an RMSE decrease of 11.32, and a SAM reduction of 0.173°. This further confirms the contribution of the proposed single-layer encoder to enhancing reconstruction performance.
2) *Effectiveness of Quantizer:* The comparison between baseline-V and Ours (8 bit) reveals that replacing V2's quantizer with V1's quantizer results in a decline in reconstruction performance while keeping bpppb nearly unchanged. Specifically, PSNR decreases by 0.41 dB, RMSE increases by 1.55, and SAM increases by 0.087°. These findings indicate that the absence of the proposed quantizer leads to a degradation in reconstruction quality. Furthermore, the comparison between baseline-III and baseline-IV shows that although bpppb increases slightly by 0.003, the reconstruction performance improves significantly. Specifically, PSNR increases by 0.43 dB, RMSE decreases by 10.37, and SAM decreases by 0.386°. This demonstrates that V2's uniform quantizer is capable of preserving more essential information for high-quality reconstruction compared to V1's quantizer.
3) *Effectiveness of Our SPHR Decoder:* The comparison between baseline-II and Ours (32-bit) reveals that replacing the proposed spatial-priority hierarchical decoder with V1's decoder significantly degrades reconstruction quality. Specifically, PSNR decreases by 2.42 dB, RMSE increases by 6.10, and SAM increases by 0.308°. This deterioration can be attributed to the decoder's workflow. In BTC-Net's decoder, feature enhancement operations are performed at a small spatial resolution, which limits its ability to capture spatial correlations among pixels effectively. Furthermore, the comparison between baseline and baseline-I shows that incorporating our SPHR decoder into V1 leads to a substantial performance improvement. Specifically, PSNR increases by 3.13 dB, RMSE decreases by 16.30, and SAM decreases by 0.407°. This enhancement is primarily due to the design of V2's encoder, where spatial upsampling is applied first. This approach allows the subsequent DR to better comprehend full-resolution spatial information and effectively fuse multiscale features within a U-shaped network. These results strongly validate the effectiveness of the proposed spatial-priority hierarchical architecture in improving reconstruction quality.

TABLE IV
ABLATION EXPERIMENTS ON COMPONENT EFFECTIVENESS OF THE PROPOSED METHOD

	FLOPs (M)	Memory (M)	Latency (ms)
V1 Encoder (32 bit)	28.87	1.83	0.84
V2 Encoder (32 bit)	13.08	1.55	0.44
V1 Decoder (32 bit)	612.66	52.70	19.97
V2 Decoder (32 bit)	2210.00	73.32	6.41
V1 Encoder (8 bit)	/	0.72	0.24
V2 Encoder (8 bit)	/	0.57	0.11

Furthermore, we evaluate the runtime efficiency of the 32- and 8-bit encoders, as well as the 32-bit decoders, for both V1 and V2. The results are summarized in Table IV. As shown in the table, the encoder of V2 exhibits improvements in both memory consumption and runtime under 32- and 8-bit settings.

For the decoder, V2 achieves significantly lower latency (6.41 ms) compared with V1 (19.97 ms). However, the decoder of V2 incurs higher memory usage (73.32 versus 52.7 MB for V1) and a larger computational cost in terms of FLOPs (2210 versus 612.66 M for V1).

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TABLE V
RESULTS OF FINE-GRAINED ABLATION ON DECODER INTERNAL MODULES (CORRESPONDING TO TABLE IV IN THIS ARTICLE)

	w/ SESAM	w/ SEFFM	PSNR↑/RMSE↓/SAM↓
①			38.99/53.60/1.880
②		✓	39.84/49.19/1.696
③	✓		39.44/50.58/1.753
④	✓	✓	**41.39/48.36/1.573**

TABLE VI
RESULTS OF DIFFERENT APPROACHES WITH VARIED ATTENTIONS IN DRAM MODULE (CORRESPONDING TO TABLE V IN THIS ARTICLE)

Approaches	PSNR↑/RMSE↓/SAM↓
w/ W-MSA [43]	36.74/70.25/2.473
w/ SW-MSA [43]	37.30/63.25/2.202
w/ Our SESAM	**41.39/48.36/1.573**

This design choice is motivated by practical deployment considerations: in real-world applications, the decoder is typically executed on ground-based platforms equipped with high-performance computing resources. Consequently, the primary design objective for the decoder is to ensure high-fidelity HSI reconstruction rather than minimizing computational overhead. Overall, this trade-off between reconstruction accuracy and computational efficiency is well aligned with the operational requirements of practical HSI processing systems.

2) Fine-Grained Ablation of Internal Modules in the Decoder: To further verify the effectiveness of the proposed SESAM, SEFFM, and DRAM modules, we perform a comprehensive ablation study targeting these three core components. Specifically, each module is sequentially removed and replaced with a common residual convolutional block that maintains the same number of parameters as the original module. This controlled experimental setup ensures that any observed performance differences can be directly attributed to the functional contribution of the replaced module, rather than variations in model capacity. The experimental results are summarized in Table V, where the best-performing metrics are highlighted in red. The key observations are as follows.

1) Replacing either the SESAM (②) or the SEFFM (③) with a residual block leads to a noticeable degradation in reconstruction quality, with the PSNR decreasing by 1.55 and 1.95 dB, respectively. This performance drop primarily stems from the inability of the common residual block to effectively model the intricate spatial and spectral correlations inherent in multiscale features.
2) Replacing the DRAM (①), which effectively removes both the SESAM and SEFFM from the architecture, results in a more severe PSNR degradation of 2.4 dB compared to the individual ablations. This suggests that the DRAM module plays a crucial role in integrating the complementary information processed by the SESAM and SEFFM, and its removal disrupts this synergistic mechanism.
3) Collectively, all three modules—SESAM, SEFFM, and DRAM—contribute significantly to enhancing reconstruction performance. This is corroborated by the fact that the model incorporating both SESAM and SEFFM (④) achieves the highest PSNR of 41.39 dB among all the ablation variants tested.

Furthermore, we conducted a comparative experiment to validate the superiority of the proposed SESAM over two widely adopted attention mechanisms: window-based multihead self-attention (W-MSA) and shifted-window multihead self-attention (SW-MSA), both of which are core components of the Swin Transformer architecture [43]. Given the extensive application of W-MSA and SW-MSA in state-of-the-art vision models, this comparative setup ensures the reliability and reference value of our evaluation results. The quantitative results of this experiment are summarized in Table VI, where the optimal performance metrics are highlighted in red. As illustrated in the table, substituting SESAM with either W-MSA or SW-MSA leads to a significant degradation in reconstruction quality: the PSNR decreases by 4.65 and 4.09 dB, respectively. This performance drop can be attributed to the inherent limitations of the two baseline attention mechanisms: For W-MSA, the attention computation is strictly confined within local windows, resulting in the complete neglect of cross-window information interaction. This constraint severely impairs the ability of the DRAM to capture global, large-scale contextual information, which is critical for high-fidelity image reconstruction. Although SW-MSA alleviates the "information isolation" issue of W-MSA by introducing window shifting, its cross-window information propagation heavily depends on the stacking depth of W-MSA/SW-MSA layers. During the downsampling process, however, the rapid reduction of spatial resolution limits the effective propagation of long-range contextual information—ultimately failing to overcome the "short-sighted" limitation in fine-grained feature enhancement required for reconstruction tasks.

E. Verification of Generalization Performance

In this section, to evaluate the generalization capability and transferability of the proposed BTC-Net V2 when applied to new hyperspectral data, a series of generalization experiments was conducted. The datasets employed in the experiments include three representative hyperspectral datasets: the AVIRIS (as introduced in Section IV-A1), WHU-Hi,[2] and Hyperion-L1R[3] datasets.

1) Verification on Aviris Dataset:

a) Dataset Setting: To validate the transferability of the proposed BTC-Net V2 on the AVIRIS dataset, we first conducted resampling on the entire dataset. Specifically, four subdatasets—Arizona, California, Modoc, and Yellowstone—were selected as the training sets. To this end,

[2]https://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm
[3]https://www.usgs.gov/centers/eros/science/usgs-eros-archive-earth-observing-one-eo-1-hyperion

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TABLE VII
COMPARISON OF GENERALIZATION CAPABILITY ON AVIRIS DATASET

Method	*Fort Myers*	*Amikougami*	*Dixie Valley*	*Dewar*	*Campbell River*	Average
	PSNR (dB) ↑ /RMSE ↓ /SAM (°) ↓					
DCSN (32 bit) [8]	33.34/113.5/2.179	30.39/68.26/1.471	34.32/173.3/1.828	29.67/65.02/1.827	33.02/45.14/2.041	32.15/93.04/1.869
BTC-Net (32 bit) [9]	36.49/79.57/1.960	33.01/38.40/1.366	33.25/171.1/2.036	32.07/38.58/1.624	35.55/28.79/2.032	34.08/71.29/1.804
CTCSN (32 bit) [27]	37.00/72.16/1.867	36.35/23.13/1.024	34.22/168.0/1.723	34.51/26.69/1.330	36.36/25.35/2.014	35.69/63.07/1.592
BTC-Net V2 (32 bit)	**38.84/61.12/1.499**	**39.41/16.55/0.753**	**37.43/164.7/1.283**	**38.23/16.80/0.878**	**37.97/20.33/1.547**	**38.37/55.91/1.192**

TABLE VIII
COMPARISON OF GENERALIZATION CAPABILITY ON WHU-HI AND HYPERION-L1R DATASETS

Method	WHU-Hi	Hyperion-L1R
	PSNR (dB) ↑ /RMSE ↓ /SAM (°) ↓	
DCSN (32 bit) [8]	29.81/0.123/3.286	32.64/82.82/1.877
BTC-Net (32 bit) [9]	30.94/0.106/3.090	33.86/70.94/1.841
CTCSN (32 bit) [27]	30.67/0.107/3.151	34.79/63.55/1.768
BTC-Net V2 (32 bit)	**32.12/0.086/2.939**	**36.63/52.65/1.500**

four models, namely DCSN [8], BTC-Net [9], CTCSN [27], and the proposed BTC-Net V2, were trained independently on these training subdatasets. Accordingly, the performance of each trained model was evaluated on five unseen subdatasets (i.e., subdatasets not involved in training): Fort Myers, Amikougami, Dixie Valley, Dewar, and Campbell River. Notably, the image acquisition scenarios corresponding to these test subdatasets were completely absent from the training process, which ensures the validity of the transferability evaluation. The detailed experimental results are presented in Table VII, where the best performance in each evaluation category is highlighted in red for clear comparison.

b) Results: From Table VII, we can find that: in each category of the test set, the proposed BTC-Net V2 achieves the best reconstruction performance, as evidenced by the highest PSNR and the lowest RMSE and SAM. Specifically, BTC-Net V2 attains an average PSNR of 38.37 dB, which is 2.68 dB higher than the second-highest PSNR of 35.69 dB achieved by CTCSN. In addition, the average RMSE of our method is 55.91, which is 7.16 lower than the second-lowest RMSE of 63.07 obtained by CTCSN. Finally, the average SAM of our method is 1.192°, which is 0.400° lower than the second-lowest SAM of 1.592° reported for CTCSN. BTC-Net V2 consistently surpasses all benchmarked methods across all test set categories, robustly demonstrating its generalization capability. This enables the model to adapt effectively to unseen data types while maintaining exceptional performance.

2) Verification on Whu-Hi Dataset:

a) Dataset Setting: To further validate the transferability of the proposed BTC-Net V2 across diverse hyperspectral datasets, we additionally evaluated its performance on the WHU-Hi dataset [44]—a well-recognized HSI dataset specifically designed for high-precision crop classification tasks. This dataset contains two high-resolution HSIs with dimensions of $270 \times 550 \times 400$ and $270 \times 940 \times 475$, respectively. Here, the value of 270 represents the number of spectral bands, which cover the wavelength range from 400 to 1000 nm. For the experimental validation on this dataset, we first cropped the original HSIs into nonoverlapping subimages with a uniform size of $135 \times 128 \times 128$, resulting in a total of 66 subimages. Subsequently, these 66 subimages were split into a training set and a testing set at a ratio of 5:1, yielding 55 subimages for the training set and 11 subimages for the testing set, respectively. The detailed comparison results of the experiments are presented in Table VIII, where the best-performing results are highlighted in red.

b) Results: As presented in Table VIII, our proposed BTC-Net V2 consistently outperforms all comparative methods across key evaluation metrics. Quantitatively, compared with BTC-Net (the second-best performing method in the comparison), BTC-Net V2 achieves a performance enhancement of +1.18 dB in PSNR, a reduction of −0.02 in RMSE, and a decrease of −0.151° in SAM. This set of results explicitly demonstrates that our proposed model exhibits excellent transferability across distinct hyperspectral datasets and practical application scenarios. More importantly, it further verifies the strong generalizability of the designed framework, which is crucial for adapting to diverse hyperspectral imaging tasks.

3) Verification on Hyperion-L1R Dataset:

a) Dataset Setting: The Hyperion sensor is mounted on the Earth Observing-1 (EO-1) satellite platform. Specifically, EO-1 is a next-generation Earth observation satellite developed by the U.S. National Aeronautics and Space Administration (NASA) for the 21st century, designed to succeed Landsat-7; it was launched in November 2000. In this study, L1R-level data of the Hyperion sensor were used to validate our proposed method. Notably, L1R-level data only contains raw digital numbers (DNs) values and has not undergone any geometric correction, which, to a certain extent, reflects the actual HSIs directly captured by the satellite sensor. Similar to the preprocessing strategy for AVIRIS data, we first removed bands that are severely affected by water vapor; additionally, negative values in the data were eliminated to avoid interference with model training. After data preprocessing, the HSI was split into 486 nonoverlapping subimages. These subimages were further divided into a

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training set and a testing set at a ratio of 9:1, resulting in 438 subimages for training and 48 subimages for testing, respectively. The comparative experimental results on the Hyperion-L1R dataset are also presented in Table VIII, where the optimal indicators are highlighted in red—consistent with the visualization standard adopted for the experiments on the WHU-Hi dataset.

b) Results: As presented in Table VIII, our proposed BTC-Net V2 outperforms all comparative methods (including DCSN, BTC-Net, and CTCSN) across three key evaluation metrics: PSNR, RMSE, and SAM. Quantitatively, compared with CTCSN—the second-best performing method in the comparison—BTC-Net V2 achieves a performance improvement of +1.84 dB in PSNR, a reduction of −10.90 in RMSE, and a decrease of −0.268° in SAM. This result strongly demonstrates that our model can adapt to challenging scenarios (e.g., HSI compression without manual preprocessing) and still maintain superior performance. Such performance further validates the promising application prospects of BTC-Net V2 in real-world satellite-based hyperspectral imaging scenarios.

V. Conclusion

In this article, we propose BTC-Net V2, an efficient learning-based HSI compression method. It consists of an ESL encoder and an SPHR decoder. The ESL encoder incorporates only one convolutional layer to compress the HSI, effectively reducing memory usage and computational pressure on satellites. The SPHR decoder restores the spatial, detailed, and spectral information sequentially to recover the full image size. This hierarchical workflow enables the decoder to better understand the spatial and spectral characteristics, resulting in high reconstruction quality. The experimental results demonstrate that our method outperforms all tested SOTA HSI compression methods, achieving a PSNR of 40.93 dB at a very low transmission bit rate of 0.064 bpppb.

References

[1] D. A. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] A. F. H. Goetz, "Three decades of hyperspectral remote sensing of the Earth: A personal view," *Remote Sens. Environ.*, vol. 113, pp. S5–S16, Sep. 2009.

[3] Y.-Z. Feng and D.-W. Sun, "Application of hyperspectral imaging in food safety inspection and control: A review," *Crit. Rev. Food Sci. Nutrition*, vol. 52, no. 11, pp. 1039–1058, Nov. 2012.

[4] X. Briottet et al., "Military applications of hyperspectral imagery," *Targets backgrounds XII: Characterization Represent.*, vol. 6239, pp. 82–89, May 2006.

[5] H. Liu, X. Su, X. Shen, and X. Zhou, "MSNet: Self-supervised multiscale network with enhanced separation training for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520313.

[6] L. Ravikanth, D. S. Jayas, N. D. G. White, P. G. Fields, and D.-W. Sun, "Extraction of spectral information from hyperspectral data and application of hyperspectral imaging for food and agricultural products," *Food Bioprocess Technol.*, vol. 10, no. 1, pp. 1–33, Jan. 2017.

[7] X. Shen et al., "Efficient hyperspectral sparse regression unmixing with multilayers," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522614.

[8] C.-C. Hsu, C.-H. Lin, C.-H. Kao, and Y.-C. Lin, "DCSN: Deep compressed sensing network for efficient hyperspectral data transmission of miniaturized satellite," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7773–7789, Sep. 2021.

[9] X. Zhou, X. Zou, X. Shen, W. Wei, X. Zhu, and H. Liu, "BTC-Net: Efficient bit-level tensor data compression network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500717.

[10] S. A. Khayam, "The discrete cosine transform (DCT): Theory and application," *Michigan State Univ.*, vol. 114, no. 1, p. 31, 2003.

[11] C. E. Heil and D. F. Walnut, "Continuous and discrete wavelet transforms," *SIAM Rev.*, vol. 31, no. 4, pp. 628–666, Dec. 1989.

[12] A. Moffat, "Huffman coding," *ACM Comput. Surveys (CSUR)*, vol. 52, no. 4, pp. 1–35, 2019.

[13] J. Rissanen and G. G. Langdon, "Arithmetic coding," *IBM J. Res. Develop.*, vol. 23, no. 2, pp. 149–162, Mar. 1979.

[14] C. Luo and J. H. McClellan, "Discrete random sampling theory," in *Proc. IEEE Int. Conf. Acoust.*, vol. 26, May 2013, pp. 5430–5434.

[15] Y. Chen, T.-Z. Huang, W. He, N. Yokoya, and X.-L. Zhao, "Hyperspectral image compressive sensing reconstruction using subspace-based nonlocal tensor ring decomposition," *IEEE Trans. Image Process.*, vol. 29, pp. 6813–6828, 2020.

[16] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

[17] A. Vaswani, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2017, pp. 6000–6010.

[18] G. K. Wallace, "The JPEG still picture compression standard," *Commun. ACM*, vol. 34, no. 4, pp. 30–44, Apr. 1991.

[19] Q. Du and J. E. Fowler, "Hyperspectral image compression using JPEG2000 and principal component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 2, pp. 201–205, Apr. 2007.

[20] A. Maćkiewicz and W. Ratajczak, "Principal components analysis (PCA)," *Comput. Geosci.*, vol. 19, no. 3, pp. 303–342, 1993.

[21] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.

[22] Y. Liu, Q. Liu, M. Zhang, Q. Yang, S. Wang, and D. Liang, "IFR-Net: Iterative feature refinement network for compressed sensing MRI," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 434–446, 2020.

[23] I. Noor and E. L. Jacobs, "Adaptive compressive sensing algorithm for video acquisition using a single-pixel camera," *J. Electron. Imag.*, vol. 22, no. 2, May 2013, Art. no. 021013.

[24] S. Yang, M. Wang, P. Li, L. Jin, B. Wu, and L. Jiao, "Compressive hyperspectral imaging via sparse tensor and nonlinear compressed sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 5943–5957, Nov. 2015.

[25] Y. Wang, L. Lin, Q. Zhao, T. Yue, D. Meng, and Y. Leung, "Compressive sensing of hyperspectral images via joint tensor tucker decomposition and weighted total variation regularization," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2457–2461, Dec. 2017.

[26] J. Zhai, S. Zhang, J. Chen, and Q. He, "Autoencoder and its various variants," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2018, pp. 415–419.

[27] L. Zhang, L. Zhang, C. Song, and P. Zhang, "Hyperspectral image compression sensing network with CNN–Transformer mixture architectures," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2704–2713.

[29] L. Wei, Z. Ma, and C. Yang, "Activation redistribution based hybrid asymmetric quantization method of neural networks," *Comput. Model. Eng. Sci.*, vol. 138, no. 1, pp. 981–1000, 2024.

[30] R. Ding et al., "Bi-SSFormer: An ultralightweight binary spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5528017.

[31] M. Amrouche, H. Carfantan, J. Idier, and V. Martin, "Statistical destriping of pushbroom-type images based on an affine detector response," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627014.

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[32] J. P. Arroyo-Mora et al., "Assessing the impact of illumination on UAV pushbroom hyperspectral imagery collected under various cloud cover conditions," *Remote Sens. Environ.*, vol. 258, Jun. 2021, Art. no. 112396.

[33] X. Ding, X. Zhang, Y. Zhou, J. Han, G. Ding, and J. Sun, "Scaling up your kernels to 31x31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11963–11975.

[34] H. Chen, X. Chu, Y. Ren, X. Zhao, and K. Huang, "PeLK: Parameter-efficient large kernel ConvNets with peripheral convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5557–5567.

[35] P. Luo, G. Xiao, X. Gao, and S. Wu, "LKD-Net: Large kernel convolution network for single image dehazing," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 1601–1606.

[36] P. Yin, J. Lyu, S. Zhang, S. Osher, Y. Qi, and J. Xin, "Understanding straight-through estimator in training activation quantized neural nets," 2019, *arXiv:1903.05662*.

[37] Y. Li, J. Hu, X. Zhao, W. Xie, and J. Li, "Hyperspectral image super-resolution using deep convolutional neural network," *Neurocomputing*, vol. 266, pp. 29–41, Nov. 2017.

[38] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[39] X. Su et al., "EigenSR: Eigenimage-bridged pre-trained RGB learners for single hyperspectral image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2025, vol. 39, no. 7, pp. 7033–7041.

[40] C. K. Huang and H. H. Nien, "Multi chaotic systems based pixel shuffle for image encryption," *Opt. Commun.*, vol. 282, no. 11, pp. 2123–2127, Jun. 2009.

[41] L. Peng, C. Zhu, and L. Bian, "U-shape transformer for underwater image enhancement," *IEEE Trans. Image Process.*, vol. 32, pp. 3066–3079, 2023.

[42] X. Liu, J. Hu, X. Chen, and C. Dong, "UDC-UNet: Under-display camera image restoration via U-shape dynamic network," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 113–129.

[43] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[44] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

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